

Lab 1 实验报告

1. 这俩咋是反过来的？

```
// The instruction decoder
CtrlUnit ctrl(
    // Inputs
    .inst      (inst_ID),           // Input instruction.
    .cmp_res   (cmp_res_ID),       // result for B type compare.
    // Outputs
    .Branch    (branch_ctrl),      // whether to move PC.
    .ALUSrc_A  (ALUSrc_A_ctrl),    // 0 for rs1 data, 1 for using PC (JAL/JALR/AUIPC).
    .ALUSrc_B  (ALUSrc_B_ctrl),    // 0 for using rs2 data, 1 for using immediate number.
    .dataToReg (dataToReg_ctrl),   // Which data as reg file input. 0 for ALU result, 1 for dataMem output.
    .regWrite  (regWrite_ctrl),    // Whether to write data back to reg file.
    .memWrite  (memWrite_ctrl),    // Whether to write data memory.
    .MIO       (MIO_ctrl),         // Debug signal, not drwan in figure.
    .ImmSel    (ImmSel_ctrl),      // To tell immediate generator how to extrace immediate from instruction.
    .cmp_ctrl  (cmp_ctrl),         // Compare type of B-type instructions, as the input of compare unit.
    .ALUControl(ALUControl_ctrl),  // ALU calculation type.
    .memAccType(memAccType_ctrl),  // Memory access size and signedness.
    .JALR      (JALR)              // Used by JALR instruction. Whether move PC to rs1+imm.
);
```

```
MUX4T1_32 mux_WB(
    .I0      (memDataOut_WB),
    .I1      (SAMVOut_WB),
    .I2      (ALUOut_WB),
    .I3      (32'd0),
    .s       (dataToReg_WB),
    .o       (rdData_WB)
);
```

最开始按照 ID_stage.v 的注释写，结果遇到某条lw指令的时候无法正确处理冲突。

2. EXE_stage.v 中 dataToReg 和 ALUOut 实际上是同步的，都只需要两个额外的寄存器来对齐。我之前以为 dataToReg 比 ALUOut 早一个周期准备好，所以多加了一个寄存器，导致一系列问题。

